

How Can Science Support Maritime Decarbonisation?

My journey from shipping to research, policy and implementation

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Today's lecture has two parts

PART 1

How I ended up in shipping research

30–40 minutes

My shipping background, an unexpected PhD decision, my interest in computers and coding, and the research I do today.

Break between the two parts · Part 2 begins around 11:00

PART 2

How science supports maritime decarbonisation

40 minutes

How evidence helps us measure emissions, compare possible pathways, and move promising ideas towards implementation.

PART 1

How I ended up in shipping research

My journey through family, shipping, computers and one unexpected decision.

Shipping came before research.

My grandfather and father both worked at sea, so shipping had always felt personal to me.

I later studied **International Shipping Management**, followed by **Logistics and Environmental Policy**—first learning how the industry works, and then asking how it could change.



A PhD was not my first choice.

After my Master's, I received an offer for a supply-chain management role at **Huawei**.

I then applied for a short Research Assistant position, mainly to help pay for my flight home.

A ten-minute conversation with Prof. Alexis Lau (**my supervisor**) changed the direction. The next day, I decided to pursue a PhD.



I simply enjoyed computer games and coding.

There was no deeper reason at the beginning: I liked spending time on a computer, playing games and writing code.

Research gave me a way to bring these interests into my daily work: working with data, building algorithms and solving difficult problems.



Today, I use data to understand how shipping works.

From the shore, we see individual ships.

In the data, we begin to see the whole system: where vessels move, where activity concentrates and where environmental impacts may occur.



The methods are complicated. The logic is simple.

STEP 1

**Observe real
activity**

STEP 2

→ **Interpret it
with code**

STEP 3

→ **Turn it into
useful evidence**

The technical details change. This basic research logic remains.

A paper is not the final destination.

The work matters when evidence enters a real conversation.

It matters even more when that conversation supports a practical next step.





The ideal plan is to stay in shipping research.

I want to keep studying how shipping works and how it can change.

Ideally, my research would influence the industry and contribute something useful to the wider world.



FROM MY STORY TO THE RESEARCH

What can science actually do for maritime decarbonisation?

After the break, we will move from my journey
to the practical role of evidence.

Part 2 begins around 11:00

Is shipping a significant source of global carbon emissions?

Before looking at the evidence, which of these views do you agree with?

- Shipping is cleaner than road transport, so its emissions are a lower priority.
- Global trade depends on ships, so strong regulation could carry economic costs.
- A fleet of more than 100,000 vessels is large enough to matter globally.
- We need an emissions estimate before deciding how strongly to regulate shipping.

How can we estimate carbon emissions from shipping?

THE BASIC CALCULATION

Fuel consumption $\times$ emission factor = CO₂ emissions

A SIMPLE EXAMPLE

1 tonne of marine diesel $\times$ 3.206¹ $\approx$ 3.2 tonnes of CO₂

The next question is therefore simple: how much fuel does the fleet use?

¹The exact emission factor depends on the fuel. Around 3.2 is a useful classroom approximation for conventional marine fuel.

But how do we know the fuel use of every ship?

At the start of 2026, the global merchant fleet included around **116,000 vessels**. We cannot collect detailed fuel records from every ship.

ONE SHIP, THEN THE WHOLE FLEET

Maritime big data gives us a moving diary for each ship.

1. See when one ship is sailing, waiting or in port.
2. Combine its activity with ship and engine information to estimate fuel use.
3. Repeat for every observed ship and add the estimates.

1 AIS MOVEMENT & SPEED OVER TIME (INPUT)



AIS OBSERVATIONS

time (UTC)	lat	lon	speed (kn)
12:00	10.00	20.00	8
12:30	10.25	20.60	8
13:00	10.70	21.50	14
...	...	...	...
14:30	12.00	24.00	8

SPEED OVER TIME



2 CUBIC RULE: POWER ~ v^3

Required power $\propto \text{speed}^3$
Power $\sim v^3$

LOW SPEED
8 kn



MEDIUM SPEED
14 kn



HIGH SPEED
18 kn



★ Doubling speed (8 → 16 kn)
increases required power $\sim 8x$

3 POWER + TECHNICAL DATA → FUEL & CO₂ PER TIME STEP

VESSEL & ENGINE INFORMATION

Main engine power (MCR)	20,000 kW
Average load factor	75 %
SFOC (at 75% load)	190 g/kWh
Fuel type	HFO
CO ₂ emission factor	3.114 tCO ₂ / t fuel

FUEL FOR EACH TIME STEP

$$\text{Fuel}_t = \text{SFOC} \times \text{Power}_t \times \Delta t$$

[kg] [g/kWh] [kW] [h]
(convert g to kg)

CO₂ FOR EACH TIME STEP

$$\text{CO}_2_t = \text{Fuel}_t \times \text{emission factor}$$

[tCO₂] [t fuel] [tCO₂ / t fuel]

SUM OVER ALL TIME STEPS FOR ONE SHIP

t ₁	t ₂	t ₃	...	t _n
Fuel ₁	Fuel ₂	Fuel ₃	...	Fuel _n
CO ₂ ₁	CO ₂ ₂	CO ₂ ₃	...	CO ₂ _n

$$\text{Total fuel (one ship)} = \sum_{i=1}^n \text{Fuel}_i$$

$$\text{Total CO}_2 \text{ (one ship)} = \sum_{i=1}^n \text{CO}_2_i$$

4 REPEAT FOR ALL SHIPS → FLEET TOTALS

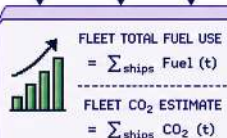


Ship A	Fuel (t)	12,345
	CO ₂ (t)	38,475

Ship B	Fuel (t)	8,210
	CO ₂ (t)	25,602

Ship C	Fuel (t)	15,876
	CO ₂ (t)	49,452

...		
Ship N	Fuel (t)	9,754
	CO ₂ (t)	30,380



FLEET TOTAL FUEL USE
= $\sum_{\text{ships}} \text{Fuel} (t)$

FLEET CO₂ ESTIMATE
= $\sum_{\text{ships}} \text{CO}_2 (t)$

A global study provides the evidence

The Fourth International Maritime Organization (IMO) Greenhouse Gas Study estimated that total shipping emitted in 2018:

**1,056 million
tonnes**
of CO₂

2.89%

of global human-caused CO₂
emissions

Science role 1: provide evidence for policy discussion

There is no single decarbonisation pathway

A transition can combine four kinds of action:

1. **Use less energy.** Improve hulls, engines and onboard systems.
2. **Operate ships differently.** Adjust speed, routing, loading and port calls.
3. **Change the energy source.** Use fuels with lower lifecycle greenhouse gas emissions.
4. **Prepare ports.** Provide bunkering, shore power, safety systems and supporting logistics.

The appropriate combination depends on the ship, route, cargo and timescale.

Implementation depends on the wider system

A promising technology still has to pass three practical tests:

1. **Can it work?**

Are compatible vessels, fuels and port infrastructure available?

2. **Can it pay?**

Can cargo demand, finance and long-term commitments support the cost?

3. **Can it operate within the rules?**

Do safety standards and public policy enable implementation?

The transition succeeds when these conditions develop together.

Green Shipping Corridors coordinate action

A Green Shipping Corridor is a route between two or more ports where partners work together to advance zero-emission shipping.

- Start with one route and an existing shipping service.
- Bring cargo owners, ship operators, ports and fuel providers into the same process.
- Align vessels, fuel, infrastructure, finance, standards and policy.



The idea is attractive. Delivery is difficult.

The 2025 global progress review tracked:

84

active Green Shipping Corridor initiatives

4

had reached the realisation stage

The feasibility wall

Many initiatives remain stalled because zero-emission fuels still cost more than conventional fuels.

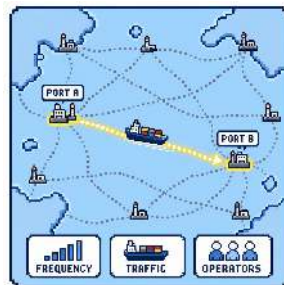
Which routes have the strongest starting conditions?

Source: Global Maritime Forum, Annual Progress Report on Green Shipping Corridors 2025.

Maritime big data helps identify promising corridors

The same movement data used to estimate emissions can reveal which routes have the strongest starting conditions.

1. Find routes with regular, repeated movement between ports.
2. Identify the vessels and operators already serving each route.
3. Compare traffic scale and service concentration across candidate routes.



Science role 2: identify routes with a credible starting point

Why San Antonio–Hong Kong stands out

2021 ROUTE ANALYSIS · #1 BY ESTIMATED FUEL DEMAND

52

two-way voyages

29

ships serving the route

23.3

average voyage days

243,725

tonnes of estimated annual
fuel demand

Why it matters: regular, long-haul traffic creates **predictable fuel demand** and a meaningful opportunity for cleaner fuels.

Research can inform policy dialogue

April 2025

I presented evidence on the route and its potential as a Green Shipping Corridor.

17 November 2025

San Antonio was announced among Hong Kong's first Partner Ports.

CONTINUING WORK

Research, stakeholder dialogue and practical coordination continue.

Hong Kong's CE unveils port partnership arrangements at ALMAC

17 November 19, 2025 | Michael Grimmer | V. Quares



Hong Kong Chief Executive John Lee (center) yesterday (17 November) unveiled a series of partnership agreements with Guangxi and Dalian ports on the mainland and San Antonio port in Chile.

What science contributes

- A more reliable understanding of the emissions problem.
 - A transparent basis for comparing possible pathways.
 - Evidence on feasibility, uncertainty and trade-offs.
 - A shared analytical foundation for coordination.
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Policy decisions also reflect institutions, economics, politics and values.

THANK YOU

Questions?

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